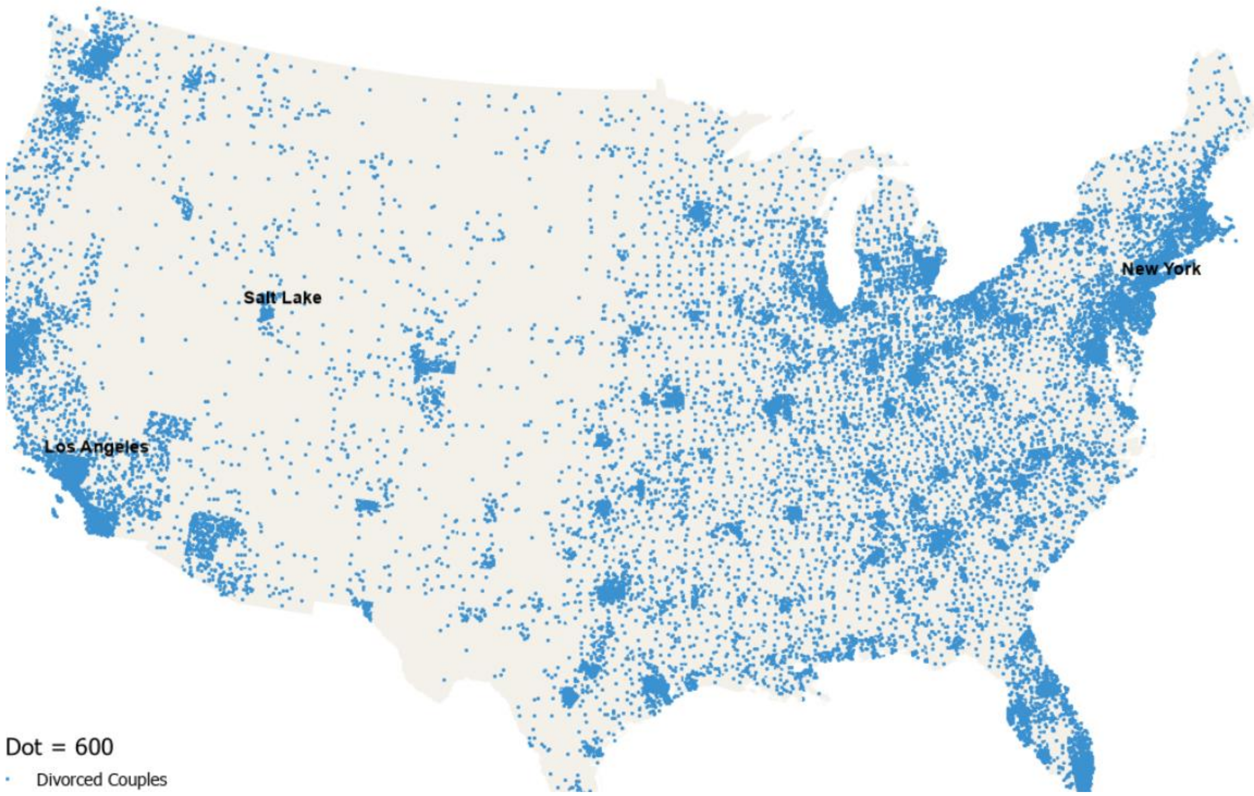
Francine Mullen

Week 5 Exercise

1. **Dot Mapping**

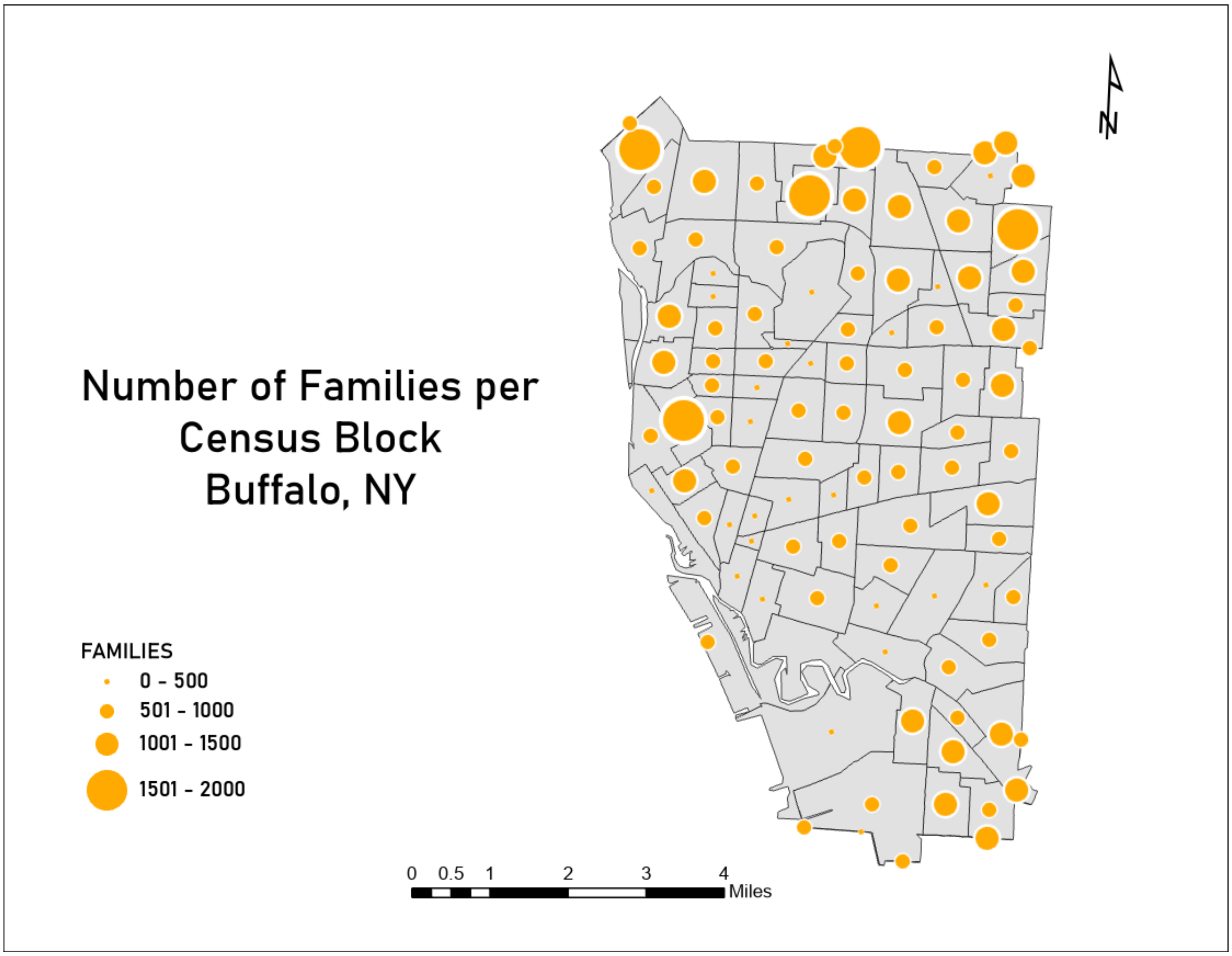


The map uses Dot Density Symbology to represent divorced couples across the United States, with each dot symbolizing 600 divorces. The highest concentrations are in major metropolitan areas like New York, Los Angeles, San Francisco, and Detroit, reflecting higher population densities and possibly varying divorce rates due to things like socio-economic factors or general statistics. Rural areas and sparsely populated regions show fewer dots, which aligns with lower overall population counts. The direct correlation can't be determined without additional information, as it statistically makes sense that higher density populations would have higher divorce rates. One rendition of this map could be comparing the population density from the Census data to the divorce rate data. The only issue I had with making this was applying the background to the polygon feature of the census tract data. I obviously got it figured out, but I for whatever reason struggled to remember how to do that.

2. Proportional Symbol Mapping

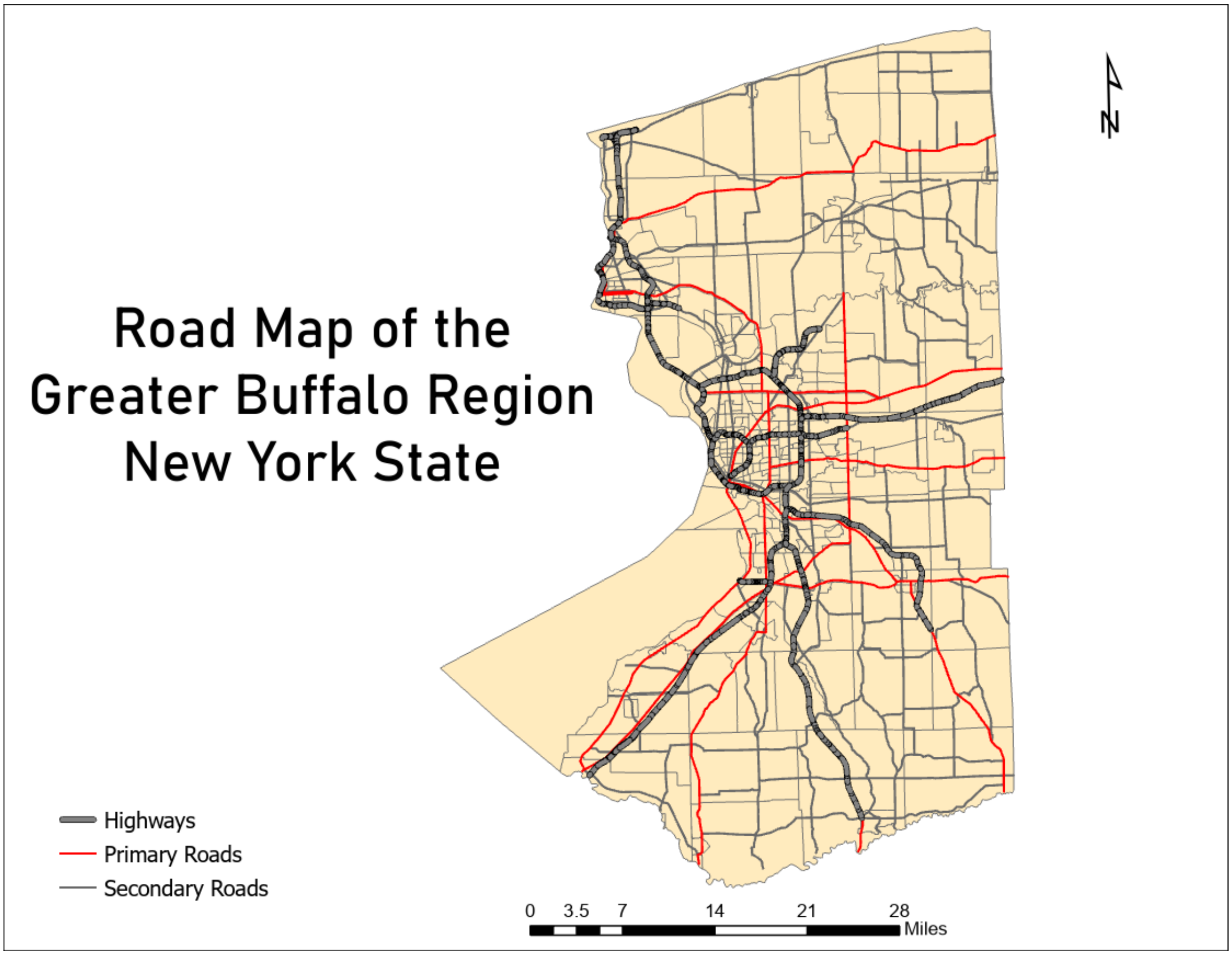
The map illustrates the population distribution of major world cities, showing a clear concentration of high-population cities in Asia, Europe, and the Americas. Cities like Beijing, New Delhi, and São Paulo have the largest circles, indicating populations exceeding 15 million. One aspect I noticed when making this is that if the size range of the circles between max and min values is too small, the larger populations can potentially represent inaccurately and visa versa. Additionally, symbol overlap was a small challenge but that was adjusted for when adjusting the range of circle size.

3. Buffalo Census Tracts



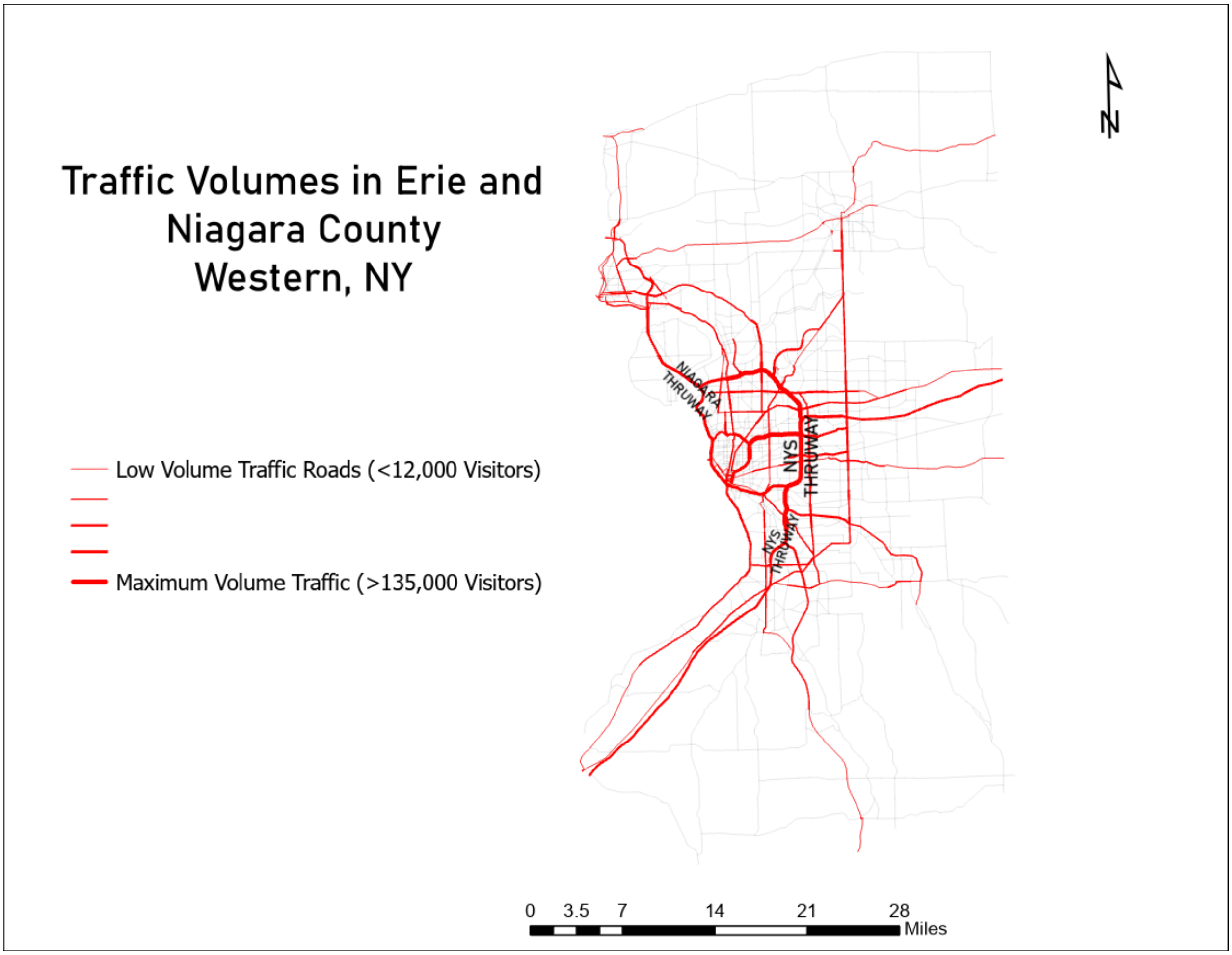
The larger circles appear more frequently in the northern and eastern parts of the city, indicating higher family concentrations, indicating that those areas are close or are a metropolitan area. While smaller circles dominate the southern and central areas, suggesting lower population densities in those census blocks. This is one map that would benefit from context like major cities or major roads. No major challenges on this one because of the exercise prior.

4. **Mapping Linear Features**



The most obvious pattern is the highways serve as major transportation corridors, connecting the city to suburban and rural areas, specifically around Buffalo which is a major city. while primary roads (red) extend outward, forming a radial pattern, which I thought was interesting. Those roads I would assume serve out of the rest of the county and connect to more suburban cities.

5. Quantitative Flow Mapping



The widest roads align with major highways, such as the New York State Thruway (I-90) and Niagara Thruway (I-190), suggesting a strong correlation between road type and traffic volume. The thinner lines represent lower-volume roads, typically secondary streets or rural routes. The high volume traffic roads do correlate with the map above matching roads that are Highways. For the legend I got creative as I didn't think it was important to the viewer to know exactly the counts of traffic. I took the approach of the range and coupled it with labeling the named roads based on the symbology.